

COMS3003A Test 3

This test contributes 10% to your final mark

Total: 20 marks

Date: 10/04/2026

1. (5 Points) Identify and correct the three errors in the following proof:

Let P be a decision problem defined as follows

$$P(\langle M \rangle) := \begin{cases} 1 & \text{if } M(\langle M \rangle) = 1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that P is decidable, therefore there is a TM M_P that decides P .

By definition, $P(\langle M \rangle) = 1 \iff M_P(\langle M \rangle) \neq 1$, therefore

$$P(\langle M_P \rangle) = 1 \iff M_P(\langle M_P \rangle) \neq 1 \quad (1)$$

By assumption, $M_P(\langle M \rangle) = 1 \iff P(\langle M \rangle) = 1$, hence

$$M_P(\langle M_P \rangle) = 1 \iff P(\langle M_P \rangle) = 1 \quad (2)$$

By Equation 1 and Equation 2,

$$M_P(\langle M_P \rangle) \neq 1 \iff M_P(\langle M_P \rangle) = 1$$

which is a contradiction.

Therefore, by contradiction, P is undecidable.

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1. *incorrect definition application error*

- The error is in Equation 1 (+1 : **Correct point of failure**)
- The equation should be that $P(\langle M_P \rangle) = 1 \iff M_P(\langle M_P \rangle) = 1$ (+1 : **Correct writing**)

2. *logical error*

- There is no contradiction (+1 : **Correct point of failure**)
- applying the correct Equation 1 has (+1 : **Correct logic**)

$$M_P(\langle M_P \rangle) = 1 \iff M_P(\langle M_P \rangle) = 1$$

3. *conclusion error*

- No contradiction means we cannot draw the final conclusion. (+1 : **conclusion does not follow**)

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1. *Change P from SELF-ACCEPT into NOT-SELF-ACCEPT (+3)*

- The language is assumedly apart of the problem statement
- They can query
- This is a single error, but stated there were 3

2. **(10 Points)** Prove that the following language is undecidable using a reduction

$$L = \{\langle M, x \rangle \mid M \text{ is in the same state for any two successive configurations of } C(M, x)\}$$

- You may assume that the following problem is undecidable

$$\text{HALT} = \{\langle M, x \rangle \mid M(x) \neq \infty\}$$

- Use the following description of the constructed Turing Machine to assist you: Modify the input TM M into M' such that it never enters the same state two configurations in a row, but is otherwise unaffected. Write the Turing Machine N onto the tape where N simulates M' on x , moves into the same state twice and then accepts. Lastly, return the result of the assumed decider on N and w .
- For the modification of M into M' , to prove that this is computable, you need to fully describe the changes made to M as part of your proof, and that these changes are computable.

RTP: HALT reduces to L **(+1 : Correct reduction)**

Assume that L is decidable, then there exists M_L that decides it.

define H on $\langle M, x \rangle$: **(+3 : Correct machine)**

- take $\langle M \rangle$ and modify it into $\langle M' \rangle$ so that it never enters the same state twice in a row, but is otherwise unaffected
 - ▶ if a state q of M transitions into itself, add a new state q' such that
 - q' has the same instructions of q that do not transition to itself,
 - if the instruction transitioned from q to itself, it now transitions into q'
 - add these same instructions for q' but move to q instead
 - ▶ **(+1: Correct modification of M into M')**
- Write $\langle N \rangle$ where N is a Turing Machine that behaves as follows:

define N on input w : **(+1: Correct N)**

$M'(x)$

Move into the same state twice

Accept

- return $M_L(\langle N, w \rangle)$ **(+1: Correct return statement $H(\langle M, x \rangle) = M_L(\langle N, w \rangle)$)**

Modifying $\langle M \rangle$ into $\langle M' \rangle$ is computable as adding a new state, and modifying instructions of known TM is computable. Writing $\langle N \rangle$ on the tape is computable as N is a well defined Turing Machine. M_L is a decider. Therefore H is a decider. **(+2: Decider)**

$$\begin{aligned} H(\langle M, x \rangle) = 1 &\iff M_L(\langle N, w \rangle) = 1 \iff \langle N, w \rangle \in L \\ &\iff N \text{ is in the same state for any two successive configurations of } C(N, w) \\ &\iff M'(x) \neq \infty \iff M(x) \neq \infty \\ &\iff \langle M, x \rangle \in \text{HALT} \end{aligned}$$

(+3: Decides HALT)

Hence, H decides HALT, but HALT is undecidable. Therefore, by contradiction, L is undecidable. **(+1: Conclusion)**

3. **(5 Points)** Prove that if L is an undecidable language, the complement $\bar{L} = \Sigma^* \setminus L$ is also undecidable

RTP: L is undecidable $\Rightarrow \bar{L}$ is undecidable

- contrapositive: $\bar{L}$ is decidable $\Rightarrow L$ is decidable

Let $\bar{L}$ be decidable, then there is a decider M for it.

define M' on x : **(+1: Correct definition of M')**

 return not $M(x)$

M is a decider, and inverting the result is decidable. Therefore M' is a decider. **(+1: M' is a decider)**

$M'(x) = 1 \iff M(x) = 0 \iff x \notin L \iff x \in \bar{L}$ hence M' decides $\bar{L}$ **(+2: M' decides $\bar{L}$)**

Therefore, by contrapositive, if L is undecidable, then $\bar{L}$ is undecidable. **(+1: Conclusion)**
